

A Comparative Study of Imputation Techniques for Clustering with Incomplete Giving Behavior Data

Christie Ngo, Baixue (Doris) Zhang, An Truong

I. Introduction

In the field of charitable giving, there are demographic trends in those who donate more money. Commonalities in characteristics, such as religion, gender, and age, may be present, forming natural groups of donation behaviors. For our project, we have chosen to investigate whether meaningful cluster structures exist within the data and whether different methods of imputation on missing data lead to different clustering results. We use Robert Mayo’s experimental data for donation behavior from a diverse population.

The dataset contains approximately 34% missing values, primarily at observation level (each observation either has all variable values or missing 6 questionnaire variable responses) in survey-related variables. Incomplete data is a common issue in real-world datasets, and imputation decisions have the potential to substantially influence downstream analyses. Therefore, it is important to thoroughly assess whether clustering results are robust to different missing data imputation methods.

To address this, we first construct a baseline dataset using complete cases (64% of total data). We then generate multiple imputed datasets using several imputation techniques, including mean/mode imputation, KNN (K-Nearest Neighbors) imputation, which uses nearest neighbors to estimate missing values [1], and MICE (Multiple Imputation by Chained Equations), which performs iterative conditional imputation based on multivariate relationships [2]. Clustering methods including, K-means, fuzzy k-means (FKM), Probabilistic Distance (PD) clustering, hierarchical clustering with an average linkage, k-prototypes, Cluster Correspondence Analysis (clusCA), skewed-t mixture model, and generalized hyperbolic mixture model, are applied consistently across all datasets.

By comparing clustering results across imputed datasets, we aim to evaluate which imputation strategies possess the most alignment with one another. Specifically, we want to assess whether different imputation methods lead to substantially different clusters and whether the underlying structure remains stable. Because we do not know the ground truth, understanding the variations between imputation techniques will help us determine which techniques would effectively produce stable outcomes.

II. Data Preprocessing

This dataset was collected from an online experiment conducted on Amazon Mechanical Turk, where participants play a modified dictator game. Each participant was given \$1 with a random multiplier and asked how much to donate. The multiplier changes how much the recipient receives, effectively altering

the price of giving. The dataset includes each participant’s decision along with demographic information such as age, income, gender, political views, and religiosity.

Data Cleaning

Each row in the dataset represents one participant who completed a Human Intelligence Task (HIT). The raw dataset contains 792 observations and 28 variables; however, several variables were removed during cleaning because they were not informative for analysis. These included identifier variables (e.g., AssignmentId, WorkerId, surveycode), variables with only a single value (e.g., Reward, HITTypeId), and redundant or overly granular variables such as location coordinates and IP address, which were already captured by the state variable. Additionally, variables like WorkTimeInSeconds and batching-related fields were excluded because they could be represented by other time and batch variables. After cleaning, the final dataset contains 14 meaningful variables, including 8 numerical variables and 6 categorical or ordinal variables, which are used for subsequent analysis. Below is a table describing each variable:

Variable Dictionary — Price of Giving / MTurk Dataset		
Variable	Type	Description
Amount	Continuous	Amount (in dollars) donated by the subject, ranging from 0 to 1.
Multiplier	Continuous	Randomly assigned multiplication factor (0–2), representing how much the receiver gets per dollar sent.
WorkTime1	Continuous	Time (in seconds) to complete the main experiment task.
WorkTime2	Continuous	Time (in seconds) to complete the follow-up survey.
Age	Ordinal	Age group of subject (e.g., 18–29, . . . , 70+).
Gender	Nominal	Self-reported gender (Male, Female, Other).
Income	Ordinal	Annual household income bracket (e.g., < 20k, 20–40k, . . . , > 100k).
Political views	Continuous	Political ideology on a 0–100 scale, where 0 = very liberal and 100 = very conservative.
Religiosity	Continuous	Self-reported religiosity on a 0–100 scale, where 0 = not religious and 100 = very religious.
Vote	Nominal	Candidate voted for in the 2016 U.S. presidential election.
State	Nominal	U.S. state of residence.
Party	Nominal	Political party self-identification, Democrat or Republican.
Attendance	Ordinal	Frequency of attending religious services (excluding weddings/funerals).
Batch	Nominal	Experimental batch indicator derived from CreationTime .

Table 1: Variable Dictionary

Feature Engineering

Several transformations were applied to improve data quality and consistency. Responses such as “prefer not to state” were recorded as missing values (NA) and later were imputed. State information was aggregated into broader geographic regions to reduce the level of categories. Income was converted into an ordinal variable to reflect its natural ordering, and gender was recorded into three categories for more consistent representation.

Missing Data

The missing data occur primarily at the observation level, meaning that each observation is either complete or missing all six questionnaire variables. Therefore, we treat the subset of fully observed data as the baseline dataset. This baseline dataset is then used to apply various clustering techniques. In total, 519 participants have complete data with no missing values.

Imputed Datasets

To address missing data, we constructed a total of five imputed datasets using different imputation methods. The variables with missing values include six questionnaire variables (e.g., political views, religious orientation, voting behavior, state, party identity, and religious attendance) as well as one timing variable (WorkTimeInSeconds), each with approximately 34.5% missingness (273 observations).

We applied three imputation approaches:

- Mean/Mode imputation, which produced one dataset
- KNN imputation, which generated two datasets using $k=5$ and $k=10$
- MICE imputation, which generated multiple datasets; from these, we randomly selected two for analysis

Imputed datasets are used to evaluate how imputation methods affect clustering results.

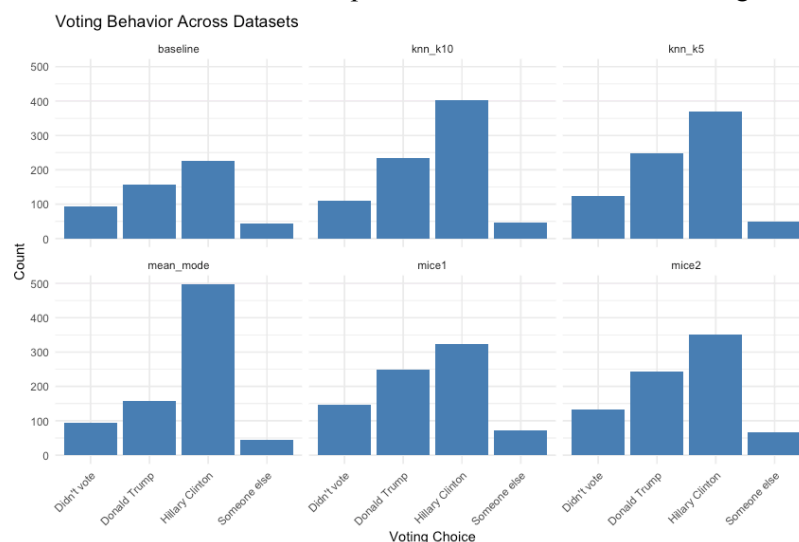


Figure 1 Distributions of Voting Across Datasets

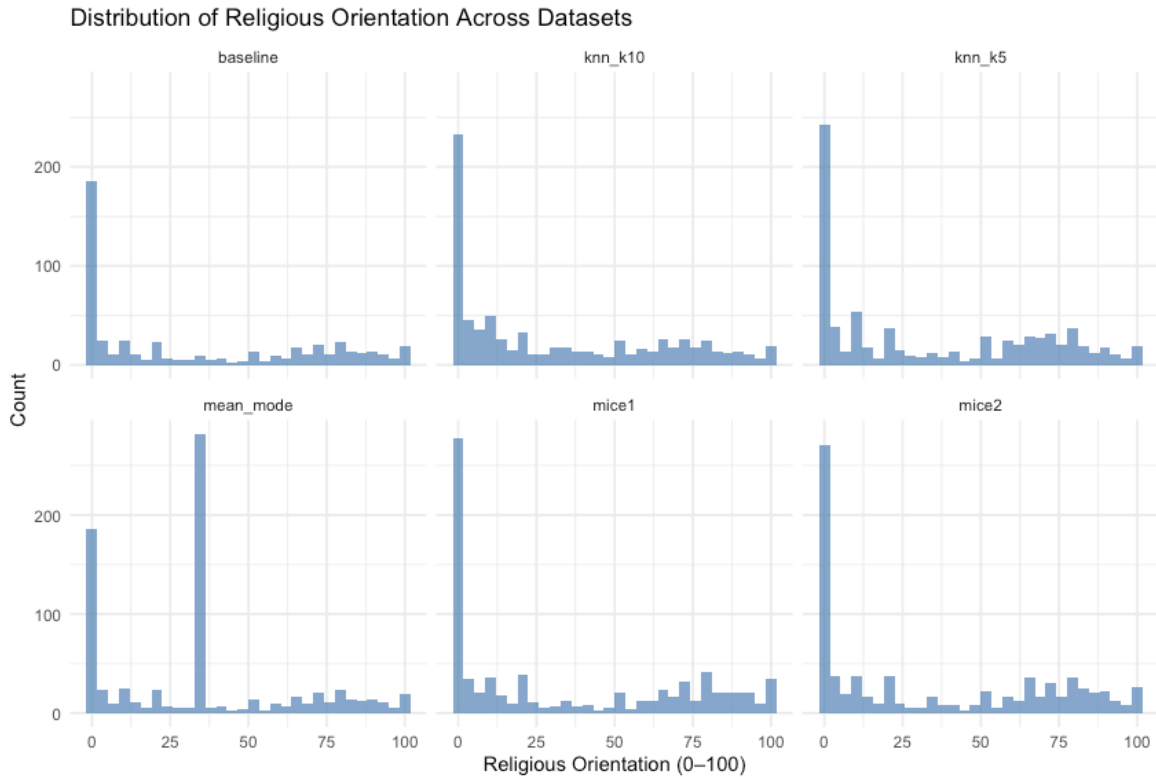


Figure 2 Distributions of Religious Orientation Across Datasets

Both categorical variables (Figure 1, showing voting choice) and numerical variables (Figure 2, showing the distribution of religious orientation) across the baseline dataset and the five imputed datasets indicate that mean/mode imputation distorts the underlying data structure, whereas KNN and MICE more effectively preserve the original distributions.

Overall, the imputed datasets exhibit relatively similar patterns. One reason is that only about 34% of the data required imputation, so the majority of the data remains unchanged. In addition, KNN and MICE produce similar estimates, resulting in comparable distributions and trends across datasets. Importantly, none of the imputation methods introduce noticeable outliers, suggesting that the overall data integrity is maintained.

After data preprocessing, we have six datasets for analysis. The first is a baseline dataset containing 519 observations, constructed using complete cases where all relevant variables are observed. We create several imputed datasets to address missing values. One dataset uses mean/mode imputation, resulting in a complete dataset with 792 observations. Two datasets are generated using K-nearest neighbors (KNN) imputation with different choices of k (e.g., $k=5$ and $k=10$). Furthermore, two datasets are constructed using Multiple Imputation by Chained Equations (MICE), reflecting the uncertainty inherent in the imputation process.

These datasets allow us to systematically compare how different imputation methods affect the structure of the data and, consequently, the clustering results. By analyzing both the baseline and imputed datasets, we aim to evaluate whether the clustering outcomes are robust to alternative missing data treatments.

III. Methods

The imputation methods include Mean/Mode imputation, K-Nearest Neighbors (KNN), and Multiple Imputation by Chained Equations (MICE). Mean/Mode imputation is the simplest method and works by imputing with the overall mean for numerical variables and mode for categorical. Although straightforward, it can underestimate variance and ignore relationships between variables. KNN imputation fills missing data with the most similar k-nearest neighbors. Because it is a distance based method, it is computationally expensive and sensitive to outliers. We selected $k = 5$ and $k = 10$ for KNN imputation, as these values provide a balance between capturing local similarity and ensuring stable estimates. Smaller k values may be sensitive to noise, while larger values can oversmooth the data and distort its structure. Our last imputation method is MICE which handles missing data by modeling each variable with missing data as a function of other variables in the dataset. In our project, MICE imputation was performed to generate multiple completed datasets. From the set of imputed datasets, we randomly selected two for further analysis.

The imputed datasets were applied to various clustering methods, including K-means, fuzzy k-means (FKM), Probabilistic Distance (PD) clustering, hierarchical clustering with an average linkage, k-prototypes, Cluster Correspondence Analysis (clusCA), skewed-t mixture model, and generalized hyperbolic mixture model. To evaluate the resulting structures, internal measures such as Dunn index, CH index, Average Silhouette Width, and the Agglomerative Coefficient were used, alongside external validation via the Adjusted Rand Index (ARI) to compare different clustering methods on agreement. Additionally, the optimal number of clusters (K) was determined using Gap statistics and Silhouette indices. For the methods with likelihood assumptions, we used BIC to select the best models.

ARI is used between each clustering method and dataset to see pairwise agreements on their respective partitions. Because our goal is to explore how different clustering results can be between imputation methods, we will only compare cluster labels on imputed datasets within the same general techniques. For example, when we explore the results using categorical columns, we will only compare the ARI between two categorical subsets using different imputation techniques.

IV. Analysis

A. Clustering Assessment

When first examining clustering tendency, the metrics that we looked into for both the complete and imputed datasets are Hopkin's statistics and using the VAT diagram. In comparing the datasets, we split them into their numerical and categorical groups. Across all the datasets, the numerical and categorical groups had high values for their Hopkin's statistics, with all of them obtaining at least 0.70 and 0.98,

respectively. The dataset with the best Hopkin’s statistic is the mean/mode imputation with 0.7980 and 0.9907 for numerical and categorical groups, respectively as well. Despite the large Hopkin’s statistics across the board, their VAT diagrams were difficult to interpret and lacked a clear clustering visualization. This made it difficult to understand how many clusters existed in the datasets as a preliminary step.

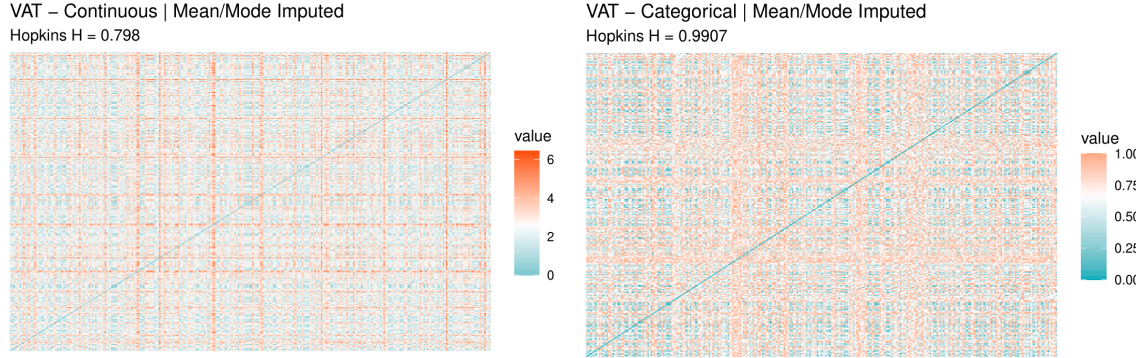


Figure 3: VAT

B. Baseline dataset with different clustering methods

We first analyzed the baseline dataset using complete cases. On standardized numerical features, we applied k-means clustering, where the silhouette method suggested $k=5$ while the elbow method favored $k=2-3$; for interpretability, we focused on $k=2$ and $k=3$. We also applied Gaussian Mixture Models (GMM), which selected $k=3$ based on BIC, and hierarchical clustering with average linkage. To incorporate categorical variables, we used k-prototypes and Gower-based hierarchical clustering on the mixed dataset. Clustering consistency across methods was evaluated using the Adjusted Rand Index (ARI).

Across baseline clustering methods, the data does not exhibit a stable clustering structure, as evidenced by very low agreement across algorithms. However, k-means ($k=2$) reveals a clear split primarily driven by religiosity. The more religious cluster tends to give more, suggesting a positive association between religiosity and prosocial behavior, although this pattern is not robust across clustering methods.

Clustering Methods for Basline Dataset - ARI

	kmeans	fuzzy_kmeans	pd_cluster	hclust_avg	gmm	kprototype
kmeans	1.0000	0.7847	0.7098	0.0041	0.0007	0.0084
fuzzy_kmeans	0.7847	1.0000	0.8560	0.0014	-0.0003	0.0032
pd_cluster	0.7098	0.8560	1.0000	-0.0002	-0.0006	0.0001
hclust_avg	0.0041	0.0014	-0.0002	1.0000	0.0443	0.3738
gmm	0.0007	-0.0003	-0.0006	0.0443	1.0000	0.1795
kprototype	0.0084	0.0032	0.0001	0.3738	0.1795	1.0000

Table 2 Clustering Methods for Baseline Dataset -ARI

C. Distance-Based Clustering Methods for Imputed Datasets

Distance-Based Methods (Numerical Data)

To obtain the optimal number of clusters k , we looked into both the gap statistic and mean silhouette for K-Means, FKM, PD Clustering, and Hierarchical Clustering with an average linkage. The optimal number of clusters per method per dataset was compared with one another using both internal and external measures. For internal measures, the mean silhouette, CH index, and the Dunn index were used to measure inter-cluster quality, compactness, and separateness with higher values indicating quality clusters. It is to note that each combination of clustering methods and datasets did not agree on the same number of clusters. The dataset with the highest values with mean silhouette and CH index is the mean/mode imputed dataset with a mean silhouette of 0.3107 for Hierarchical Average ($k = 2$) and CH index of 257.0552 for K-Means ($k = 5$). Since Hierarchical clustering is the only method with an agglomerative coefficient, it cannot be compared with the other methods. Nonetheless, we obtained the highest agglomerative coefficient of 0.9171 from the mean/mode imputed dataset as well.

Method	K	Silhouette	CH_Index	Dunn_Index	Agg_Coef
K-Means	5	0.2882	257.0552	0.0140	
Fuzzy K-Means	4	0.2681	231.0384	0.0112	
PD Clustering	2	0.2072	145.0197	0.0103	
Hierarchical Avg	2	0.3107	170.0892	0.1011	0.9171

Table 3 Indices for Different Clustering Methods on Numerical Data

The method and dataset that had the highest Dunn index is 0.0972 for the Hierarchical clustering ($k = 3$) for the complete and complete baseline datasets (see Appendix). As for external measures, we performed pairwise comparisons on ARI between each model and dataset. It is imperative to note that since we are using the optimal number of clusters K specific to each method and respective dataset, low ARI values may indicate different K choices rather than disagreements in cluster structure. Thus the use of ARI in this case is not necessarily an external validation measure but one that measures partition similarity rather than agreement. Interpreting higher ARI suggests similarities in partition structure whereas lower values may reflect scale differences rather than structure disagreements. Despite the varying number of clusters per method, K-Means and FKM had the highest ARI values for every imputed dataset with the highest being in KNN ($k = 5$) imputation with a value of 0.8162 (see Appendix).

ARI Comparison Across Imputed Datasets

Comparison	K-means	Fuzzy K-means	Partitioning	Hierarchical
Mean/Mode vs MICE 1	0.5102	0.1891	0.1221	0.0610
Mean/Mode vs MICE 2	0.4428	0.4013	0.1348	0.3143
Mean/Mode vs KNN k=5	0.4516	0.4088	0.1561	0.6833
Mean/Mode vs KNN k=10	0.7491	0.6671	0.3665	0.8385
MICE 1 vs MICE 2	0.4430	0.1116	0.4957	0.3666
MICE 1 vs KNN k=5	0.4451	0.1037	0.4993	0.0734
MICE 1 vs KNN k=10	0.5443	0.1865	0.4074	0.0457
MICE 2 vs KNN k=5	0.5055	0.4722	0.5174	0.2581
MICE 2 vs KNN k=10	0.4556	0.3602	0.4172	0.2858
KNN k=5 vs KNN k=10	0.5407	0.3805	0.5927	0.6392

Table 4 ARI Comparison Across Imputed Datasets

As shown in Table 5, the different imputed datasets exhibit unstable clustering within the same clustering method, as indicated by the moderate and highly variable ARI values across imputation pairs. While a few combinations (e.g., KNN k=5 vs k=10 or mean/mode vs KNN k=10) show relatively higher agreement, many others remain only weakly consistent. This suggests that clustering results are sensitive to the choice of imputation method and that the underlying cluster structure is not strongly defined.

Distance-Based Methods (Categorical Data)

Because there are several categorical variables that pertain to valuable demographic information, such as age, income, political party, and religious participation, we explored Cluster Correspondence Analysis (clusCA) on this subset of variables. Using the “elbow method” from a Multiple Correspondence Analysis (MCA) scree plot, we can see the amount of variance explained by increasing the number of components tapers off after 2-3 dimensions, providing us a range to test. Based on the other methods and dataset size, we limit our exploration from 2 to 8 clusters. We chose the best model for each dataset using the Average Silhouette Width (ASW). This is an ideal metric to balance both cohesion of points within the same cluster and separation between different clusters.

Across the different imputed datasets, we can see that all selected 4 clusters and had similar ASW scores. For our categorical variables, the clustering structure is consistent but the way points were assigned to each cluster varies. The ARI table below highlights that the cluster labels have low alignment between different imputation techniques. Even between the two MICE datasets, there is a mismatch in more than half of the rows (ARI = 0.471). The highest proportion of concordance is between KNN-5 and KNN-10 with an ARI of 0.708. This inconsistency is likely due to the discrete nature of these variables. Because there are only a few possible factors, MICE and KNN can both quickly increase the noisiness of the data in different ways whereas Mean/Mode imputation greatly reduces variance. We opted out of tuning different parameters to increase the complexity of this method since the results seem quite stable, and we are unable to definitively choose the best method without ground truth.

Categorical Dataset	K	ASW
Baseline	4	0.292
MM	4	0.376
Mice 1	4	0.289
Mice 2	4	0.288
KNN-5	4	0.309
KNN-10	4	0.321

Table 5 ASK Across Different Datasets for Categorical Data

Categorical Dataset	MM	Mice 1	Mice 2	KNN-5	KNN-10
MM	1.000	0.232	0.279	0.319	0.410
Mice 1	0.232	1.000	0.471	0.457	0.410
Mice 2	0.279	0.471	1.000	0.481	0.487
KNN-5	0.319	0.457	0.481	1.000	0.708
KNN-10	0.410	0.410	0.487	0.708	1.000

Table 6 ARI for Different Datasets Using MCA

Distance-Based Methods (Mixed Data)

We applied k-prototypes clustering to all six datasets. For the baseline dataset, the silhouette method suggested two clusters, while the elbow method indicated that two or three clusters were more appropriate. Similar patterns were observed across all imputed datasets (mean/mode, KNN with k=5 and k=10, and MICE), where the silhouette method consistently suggested two clusters and the elbow method suggested two or three clusters. Based on these results, we focused on k=2 and k=3 for further analysis. The clustering results are broadly consistent across datasets, as shown by the ARI comparison in Table 7:

	knn_k5	knn_k10	mice1	mice2	mean_mode
knn_k5	1.0000	0.9444	0.7144	0.6294	0.9444
knn_k10	0.9444	1.0000	0.6856	0.6397	1.0000
mice1	0.7144	0.6856	1.0000	0.5713	0.6856
mice2	0.6294	0.6397	0.5713	1.0000	0.6397
mean_mode	0.9444	1.0000	0.6856	0.6397	1.0000

Table 7 ARI Results for K Prototypes Clustering Across Imputed Datasets

Table 8 presents the mean values of key variables with two clusters under different imputation methods. Across all imputation methods, clusters are primarily distinguished by religiosity, which strongly correlates with political ideology and voting behavior. The more religious group tends to be more conservative, more supportive of Trump, slightly younger, and marginally more generous. In contrast, income and gender show minimal differences across clusters, suggesting they do not drive the clustering.

Dataset	n	Cluster 1							Cluster 2						
		Amt	Pol	Relig	Age	Inc	Female	Trump	Amt	Pol	Relig	Age	Inc	Female	Trump
KNN k=5	44 / 748	0.367	69.6	39.2	1.73	3.32	0.364	0.409	0.330	63.9	34.0	2.19	3.26	0.477	0.309
KNN k=10	40 / 752	0.386	68.9	37.1	1.78	3.28	0.400	0.375	0.329	64.5	30.6	2.18	3.26	0.475	0.290
MICE 1	69 / 723	0.302	64.9	36.2	1.77	3.29	0.406	0.333	0.335	62.0	34.2	2.20	3.26	0.477	0.313
MICE 2	75 / 717	0.340	68.3	45.7	1.73	3.24	0.400	0.373	0.332	65.4	33.1	2.21	3.26	0.478	0.298
Mean/Mode	40 / 752	0.386	68.8	36.4	1.78	3.25	0.400	0.350	0.329	63.8	33.4	2.18	3.25	0.475	0.190

Table 8 K Prototypes: Cluster Summary by Imputed Datasets

D. Model-Based Partitioning for Imputed Datasets

Though our data exhibits strong skewness in some variables, we were interested in learning how model-based techniques performed across the imputed datasets. These methods assume a probability distribution for the clusters in the data and iteratively update the parameters to maximize the likelihood of observing this data. Based on the skewed distributions of several covariates, we chose to fit a skewed-t and a generalized hyperbolic mixture model. Because these techniques are for continuous data, we chose to compare performances on a purely continuous dataset as well as a dataset that added ordinal variables. Ordinal variables were created by converting variables into factors with ordered integer levels. The range of clusters we tested for each model was based on trial-and-error. At higher cluster numbers, the parameter estimates were unstable, and they sometimes failed to converge entirely.

Across both versions of each dataset, the number of clusters selected for the skewed-t mixture models remained constant. We opt to use BIC as it prevents the model from overfitting and becoming overly complex. The BIC increased for the version that included ordinal variables; this is expected as BIC will always increase with more variables. The skewed-t failed to converge for the mean/mode imputed dataset. After considering the statistical motivation for skewed-t mixture models, this makes sense as the artificially reduced center variance from this particular imputation method contradicts the algorithm's purpose to fit heavy tails for outliers. Since the generalized hyperbolic distribution is more flexible, each dataset was fitted successfully. Clusters ranged from 5 to 8 with BICs fairly consistent between datasets. Again, mean/mode imputation is an outlier with 3 clusters performing the best for the continuous dataset.

Continuous Dataset	Skew t G	Skew t BIC	GHD G	GHD BIC
Baseline	2	6761.286	6	-3377.189
MM	FAILED	N/A	3	294.444
Mice 1	2	9652.748	5	-4511.356
Mice 2	2	10278.560	7	-4744.835
KNN-5	2	10393.910	6	-4606.135
KNN-10	2	9718.209	8	-4035.763

Table 9. Continuous Data Cluster Results (*Let G stand for the number of clusters*)

Continuous & Ordinal Dataset	Skew t G	Skew t BIC	GHD G	GHD BIC
Baseline	2	9854.426	7	-7698.030
MM	FAILED	N/A	6	5146.632
Mice 1	2	15026.130	5	-10253.890
Mice 2	2	14934.010	5	-10346.890
KNN-5	2	15017.320	8	-10061.460
KNN-10	3	14451.850	5	-10397.470

Table 10. Continuous & Ordinal Data Cluster Results (*Let G stand for the number of clusters*)

Interestingly, in the purely continuous datasets, the ARI values are much higher between MICE and KNN imputed datasets when comparing within each mixture distribution. Mean/Mode imputation consistently differed on cluster labels with the rest of the other datasets. The highest ARI method across imputation techniques was for the KNN-5 with the first MICE dataset at 0.727 from the generalized hyperbolic model results. These ARI values were much lower when we combined the dataset with some ordinal variables. We believe this is reasonable as the ordinal variables contributed to more variance for the distributions to model. Treating the categories of ordinal variables as equally-spaced also leads to more misrepresented relationships. The full ARI comparison table is in the appendix.

E. Cluster Demographic Exploration

Additionally, we are interested in seeing if the clusters observed reflect the same trends as the author’s regression findings.

To do so, we first used the k-prototypes method, which allows us to incorporate both numerical and categorical variables and thus capture the full structure of the dataset. The k-prototypes clustering reveals a more interpretable segmentation compared to purely distance-based methods. In particular, the clusters are meaningfully differentiated along key demographic and behavioral dimensions, as shown in table #. One cluster tends to exhibit higher levels of religiosity, more conservative political views, younger participants, and a greater likelihood of supporting Trump, while the other cluster is characterized by lower religiosity and more liberal political orientation. The more religious group tends to donate slightly more on average, suggesting a potential link between religiosity and prosocial behavior. However, demographic variables such as income and gender do not appear to strongly differentiate the clusters, indicating that these factors are not primary drivers of the segmentation.

Overall, the k-prototype results reinforce the idea that religiosity serves as an organizing dimension in the data, with the more religious group also exhibiting more conservative political views, younger, higher support for Trump, and slightly higher giving behavior as shown in Table 8.

We then dive into the least complex model-based method, skewed-t. Because we may have outliers and differing cluster sizes, we believe that previous distance-based methods may not be as robust. We first

examine the two clusters obtained by the skew-t distribution using the continuous KNN-5 imputed dataset. Surprisingly, the distinguishing feature between the two clusters is not any of the demographic information. The political and religious views were similar in these groupings, but they were notably different in the amount of time taken to answer the two surveys. The time participants spent on the survey was one of the most skewed variables, so it makes sense that the skewed-t model may have compensated for this.

Cluster	N	Mean Scores (Continuous)						Proportions (Categorical)			
		Age	Income	Politics	Religion	Time 1	Time 7	Female	Trump	Never Att.	Democrat
Cluster 1	268	1.94	3.14	68.50	32.90	525.0	498.0	0.358	0.369	0.511	0.541
Cluster 2	524	2.28	3.33	62.80	29.90	125.0	60.8	0.529	0.256	0.584	0.542

Table 11. Overall Clusters Based on Continuous Clustering (Skewed-t mixture model)

We then observed the demographic information for the continuous + ordinal dataset using the best mixture model under a skewed-t distribution as well. These results belonged to the MICE imputation technique. Once we introduce the ordinal variables, the clusters seem to be much more informative. There are now strong separations based on religious ideology along with political outcomes (party affiliation and presidential vote in 2016). One cluster is much more religious and Republican-dominant while the second is the opposite.

Cluster	N	Mean Scores (Continuous/Ordinal)							Proportions (Categorical)			
		Age	Amount	Income	Politics	Religion	Time 1	Time 7	Female	Trump	Never Att.	Democrat
Cluster 1	458	2.14	0.359	3.22	57.90	54.20	363.0	450.0	0.459	0.400	0.273	0.362
Cluster 2	334	2.20	0.296	3.31	68.10	7.20	119.0	59.0	0.488	0.198	0.698	0.533

Table 12. Overall Clusters Based on Continuous + Ordinal Clustering (Skewed-t mixture model)

Furthermore, analysis of the clusters reveals that more religious individuals tend to give more, which is consistent with the exploratory experiment findings reported in the original paper.

V. Conclusion & Summary

Ultimately, we recognize that the imputation technique chosen will have a range of impact on the clustering results, and the extent of the variation depends on several factors.

From the distribution of the imputed datasets, we observed mean/mode imputation distorts both categorical and numerical distributions, while KNN and MICE better preserve the original data structure. Overall patterns remain consistent across datasets because most data is unchanged (~66%), and KNN and MICE produce similar estimates without introducing outliers.

The baseline dataset analysis shows that the clustering structure differs substantially across methods. This is expected because different algorithms rely on distinct assumptions about cluster shape, distance metrics, and data structure. Methods such as k-means, fuzzy k-means, and partitioning around medoids

(pd_cluster) show relatively higher agreement, while hierarchical clustering, GMM, and k-prototypes produce markedly different partitions. The very low ARI values between many method pairs suggest that the underlying cluster signal is weak and not well-defined, making the results highly sensitive to the choice of algorithm.

For the distance-based clustering methods on numerical data, the best imputation for these methods based on the internal validation metrics is the mean/mode imputed dataset. This result is surprising because it is the most simple compared to the other imputation methods. This could suggest that the data lacks structure for clustering, and this is evident in the mean silhouette scores across the board for all methods being close to 0.20-0.30. MICE and KNN imputation assume a correlation structure, hence were not fitting options for our dataset. In summary, mean/mode is not universally better than the other methods but rather for this specific data and its missingness patterns.

In the limited subset of only categorical variables, we can see that the imputation method did not result in drastically different clustering for MCA. Because there are a limited number of categories for the technique to assign observations to, the values selected based on probability or similarity could oftentimes be the mode.

Notably, the k-prototypes results are highly consistent across different imputed datasets, as reflected by the relatively high ARI values in Table 7. This suggests that the clustering structure is robust to the choice of imputation method with k-prototypes clustering, with similar patterns observed across KNN, MICE, and mean/mode datasets.

When considering the model-based techniques for our skewed data, we can see that Mean/Mode imputation is notably ineffective at revealing the cluster structures. This method of imputing missing values reinforces the central signal and inhibits the modeling techniques from properly capturing the true data variations. Without the ground truth, we cannot definitively say which alternative, KNN or MICE, is better for our data. KNN is simpler to compute but relies more on similar “neighboring” behaviors whereas MICE can maintain complicated interactions between covariates. Based on the background of this dataset, KNN likely performs superiorly as the demographic separations seem clearly defined. We would favor MICE if we had a larger dataset with potential to reveal more nuanced groupings.

Our data did indeed have a clustering structure across all clustering techniques we tested, but we are unable to determine the best results without the ground truth. Using the demographic information within cluster results, we did obtain similar findings to the author of the dataset, especially regarding the distinction between donation behaviors based on religiosity. For future works, we believe that it is worth exploring MissForest, a non-parametric technique, to strengthen robustness to outliers and remove a linear relationship assumption. Based on our extensive results, it would be equally compelling to explore the stability of how observations get clustered; pinpointing members who almost always get grouped together across techniques would reveal significant core groups of demographics within our data. With more observations, we would also then be able to explore behaviors for the group of participants who opted to not answer certain questions.

VI. References

- [1] S. van Buuren and K. Groothuis-Oudshoorn, “mice: Multivariate imputation by chained equations in R,” *Journal of Statistical Software*, vol. 45, no. 3, pp. 1–67, 2011.
- [2] O. Troyanskaya, M. Cantor, G. Sherlock, P. Brown, T. Hastie, R. Tibshirani, D. Botstein, and R. B. Altman, “Missing value estimation methods for DNA microarrays,” *Bioinformatics*, vol. 17, no. 6, pp. 520–525, 2001.

* Assistance with troubleshooting R code from ChatGPT and Claude LLMs

VII. Appendix

Internal Validation Summary Across All Datasets

Dataset	Method	K	Silhouette	CH_Index	Dunn_Index	Agg_Coef
Complete Data	K-Means	8	0.2850	162.2568	0.0294	
Complete Data	Fuzzy K-Means	4	0.2613	164.2588	0.0175	
Complete Data	PD Clustering	2	0.2453	65.7729	0.0251	
Complete Data	Hierarchical Avg	3	0.2593	149.4789	0.0972	0.8815
Complete Baseline	K-Means	8	0.2850	162.2568	0.0294	
Complete Baseline	Fuzzy K-Means	4	0.2613	164.2588	0.0175	
Complete Baseline	PD Clustering	2	0.2453	65.7729	0.0251	
Complete Baseline	Hierarchical Avg	3	0.2593	149.4789	0.0972	0.8815
Mean/Mode Imputed	K-Means	5	0.2882	257.0552	0.0140	
Mean/Mode Imputed	Fuzzy K-Means	4	0.2681	231.0384	0.0112	
Mean/Mode Imputed	PD Clustering	2	0.2072	145.0197	0.0103	
Mean/Mode Imputed	Hierarchical Avg	2	0.3107	170.0892	0.1011	0.9171
MICE Imputed 1	K-Means	7	0.2815	243.5778	0.0206	
MICE Imputed 1	Fuzzy K-Means	2	0.2548	242.0077	0.0442	
MICE Imputed 1	PD Clustering	2	0.2406	119.7539	0.0222	
MICE Imputed 1	Hierarchical Avg	3	0.2473	224.9475	0.0604	0.8969
MICE Imputed 2	K-Means	8	0.2824	243.9648	0.0227	
MICE Imputed 2	Fuzzy K-Means	8	0.2724	210.5918	0.0152	
MICE Imputed 2	PD Clustering	2	0.2390	106.6594	0.0295	
MICE Imputed 2	Hierarchical Avg	3	0.2493	204.6626	0.0933	0.8967
KNN k=5	K-Means	8	0.2906	247.1376	0.0275	
KNN k=5	Fuzzy K-Means	8	0.2789	218.9233	0.0157	
KNN k=5	PD Clustering	2	0.2391	104.5658	0.0115	
KNN k=5	Hierarchical Avg	2	0.2481	147.3748	0.0936	0.8977
KNN k=10	K-Means	6	0.2805	250.8733	0.0193	
KNN k=10	Fuzzy K-Means	4	0.2694	248.4977	0.0166	
KNN k=10	PD Clustering	2	0.2336	121.4763	0.0262	
KNN k=10	Hierarchical Avg	2	0.2660	161.5770	0.0843	0.8978

Pairwise ARI Summary Across All Datasets

Dataset	Comparison	ARI
Complete Data	K-Means vs Fuzzy K-Means	0.3922
Complete Data	K-Means vs PD Clustering	0.1991
Complete Data	K-Means vs Hierarchical	0.2966
Complete Data	Fuzzy K-Means vs PD Clustering	0.2994
Complete Data	Fuzzy K-Means vs Hierarchical	0.3341
Complete Data	PD Clustering vs Hierarchical	0.4371
Complete Baseline	K-Means vs Fuzzy K-Means	0.3922
Complete Baseline	K-Means vs PD Clustering	0.1991
Complete Baseline	K-Means vs Hierarchical	0.2966
Complete Baseline	Fuzzy K-Means vs PD Clustering	0.2994
Complete Baseline	Fuzzy K-Means vs Hierarchical	0.3341
Complete Baseline	PD Clustering vs Hierarchical	0.4371
Mean/Mode Imputed	K-Means vs Fuzzy K-Means	0.7860
Mean/Mode Imputed	K-Means vs PD Clustering	0.2793
Mean/Mode Imputed	K-Means vs Hierarchical	0.1199
Mean/Mode Imputed	Fuzzy K-Means vs PD Clustering	0.3298
Mean/Mode Imputed	Fuzzy K-Means vs Hierarchical	0.0982
Mean/Mode Imputed	PD Clustering vs Hierarchical	-0.0066
MICE Imputed 1	K-Means vs Fuzzy K-Means	0.2285
MICE Imputed 1	K-Means vs PD Clustering	0.2211
MICE Imputed 1	K-Means vs Hierarchical	0.3765
MICE Imputed 1	Fuzzy K-Means vs PD Clustering	0.7631
MICE Imputed 1	Fuzzy K-Means vs Hierarchical	0.4460
MICE Imputed 1	PD Clustering vs Hierarchical	0.4363
MICE Imputed 2	K-Means vs Fuzzy K-Means	0.8106
MICE Imputed 2	K-Means vs PD Clustering	0.1856
MICE Imputed 2	K-Means vs Hierarchical	0.1763
MICE Imputed 2	Fuzzy K-Means vs PD Clustering	0.1802
MICE Imputed 2	Fuzzy K-Means vs Hierarchical	0.1769
MICE Imputed 2	PD Clustering vs Hierarchical	0.1853
KNN k=5	K-Means vs Fuzzy K-Means	0.8162
KNN k=5	K-Means vs PD Clustering	0.1988
KNN k=5	K-Means vs Hierarchical	0.0590
KNN k=5	Fuzzy K-Means vs PD Clustering	0.1892
KNN k=5	Fuzzy K-Means vs Hierarchical	0.0506
KNN k=5	PD Clustering vs Hierarchical	0.0496
KNN k=10	K-Means vs Fuzzy K-Means	0.6937
KNN k=10	K-Means vs PD Clustering	0.2304
KNN k=10	K-Means vs Hierarchical	0.1034
KNN k=10	Fuzzy K-Means vs PD Clustering	0.2956
KNN k=10	Fuzzy K-Means vs Hierarchical	0.1068
KNN k=10	PD Clustering vs Hierarchical	0.0197

Cross-Dataset ARI Comparisons(K-Means, Fuzzy K-Means, PD Clustering, Hierarchical)

Comparison	ARI
Complete Data vs Complete Baseline	1.0000
Complete Data vs Mean/Mode Imputed	-0.0038
Complete Data vs MICE Imputed 1	-0.0022
Complete Data vs MICE Imputed 2	-0.0008
Complete Data vs KNN k=5	-0.0004
Complete Data vs KNN k=10	-0.0014
Complete Baseline vs Mean/Mode Imputed	-0.0038
Complete Baseline vs MICE Imputed 1	-0.0022
Complete Baseline vs MICE Imputed 2	-0.0008
Complete Baseline vs KNN k=5	-0.0004
Complete Baseline vs KNN k=10	-0.0014
Mean/Mode Imputed vs MICE Imputed 1	0.5102
Mean/Mode Imputed vs MICE Imputed 2	0.4428
Mean/Mode Imputed vs KNN k=5	0.4516
Mean/Mode Imputed vs KNN k=10	0.7491
MICE Imputed 1 vs MICE Imputed 2	0.4430
MICE Imputed 1 vs KNN k=5	0.4451
MICE Imputed 1 vs KNN k=10	0.5443
MICE Imputed 2 vs KNN k=5	0.5055
MICE Imputed 2 vs KNN k=10	0.4556
KNN k=5 vs KNN k=10	0.5407

Comparison	ARI
Complete Data vs Complete Baseline	1.0000
Complete Data vs Mean/Mode Imputed	-0.0011
Complete Data vs MICE Imputed 1	-0.0013
Complete Data vs MICE Imputed 2	0.0041
Complete Data vs KNN k=5	0.0039
Complete Data vs KNN k=10	-0.0006
Complete Baseline vs Mean/Mode Imputed	-0.0011
Complete Baseline vs MICE Imputed 1	-0.0013
Complete Baseline vs MICE Imputed 2	0.0041
Complete Baseline vs KNN k=5	0.0039
Complete Baseline vs KNN k=10	-0.0006
Mean/Mode Imputed vs MICE Imputed 1	0.1891
Mean/Mode Imputed vs MICE Imputed 2	0.4013
Mean/Mode Imputed vs KNN k=5	0.4088
Mean/Mode Imputed vs KNN k=10	0.6671
MICE Imputed 1 vs MICE Imputed 2	0.1116
MICE Imputed 1 vs KNN k=5	0.1037
MICE Imputed 1 vs KNN k=10	0.1865
MICE Imputed 2 vs KNN k=5	0.4722
MICE Imputed 2 vs KNN k=10	0.3602
KNN k=5 vs KNN k=10	0.3805

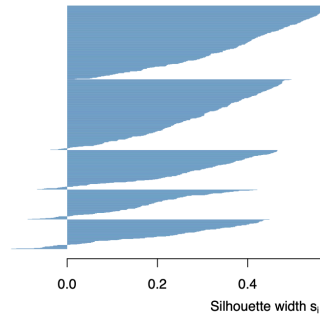
Comparison	ARI
Complete Data vs Complete Baseline	1.0000
Complete Data vs Mean/Mode Imputed	0.0008
Complete Data vs MICE Imputed 1	-0.0006
Complete Data vs MICE Imputed 2	-0.0019
Complete Data vs KNN k=5	-0.0018
Complete Data vs KNN k=10	-0.0019
Complete Baseline vs Mean/Mode Imputed	0.0008
Complete Baseline vs MICE Imputed 1	-0.0006
Complete Baseline vs MICE Imputed 2	-0.0019
Complete Baseline vs KNN k=5	-0.0018
Complete Baseline vs KNN k=10	-0.0019
Mean/Mode Imputed vs MICE Imputed 1	0.1221
Mean/Mode Imputed vs MICE Imputed 2	0.1348
Mean/Mode Imputed vs KNN k=5	0.1561
Mean/Mode Imputed vs KNN k=10	0.3665
MICE Imputed 1 vs MICE Imputed 2	0.4957
MICE Imputed 1 vs KNN k=5	0.4993
MICE Imputed 1 vs KNN k=10	0.4074
MICE Imputed 2 vs KNN k=5	0.5174
MICE Imputed 2 vs KNN k=10	0.4172
KNN k=5 vs KNN k=10	0.5927

Comparison	ARI
Complete Data vs Complete Baseline	1.0000
Complete Data vs Mean/Mode Imputed	-0.0066
Complete Data vs MICE Imputed 1	-0.0047
Complete Data vs MICE Imputed 2	-0.0031
Complete Data vs KNN k=5	-0.0098
Complete Data vs KNN k=10	-0.0117
Complete Baseline vs Mean/Mode Imputed	-0.0066
Complete Baseline vs MICE Imputed 1	-0.0047
Complete Baseline vs MICE Imputed 2	-0.0031
Complete Baseline vs KNN k=5	-0.0098
Complete Baseline vs KNN k=10	-0.0117
Mean/Mode Imputed vs MICE Imputed 1	0.0610
Mean/Mode Imputed vs MICE Imputed 2	0.3143
Mean/Mode Imputed vs KNN k=5	0.6833
Mean/Mode Imputed vs KNN k=10	0.8385
MICE Imputed 1 vs MICE Imputed 2	0.3666
MICE Imputed 1 vs KNN k=5	0.0734
MICE Imputed 1 vs KNN k=10	0.0457
MICE Imputed 2 vs KNN k=5	0.2581
MICE Imputed 2 vs KNN k=10	0.2858
KNN k=5 vs KNN k=10	0.6392

Distance-based Clustering Methods with Optimal Value k for Mean/Mode Imputed Dataset

K-Means Silhouette | k=5 | Mean/Mode Imputed

n = 792



5 clusters C_j

$j : n_j | \text{ave}_{i \in C_j} s_i$

1 : 240 | 0.37

2 : 230 | 0.29

3 : 129 | 0.27

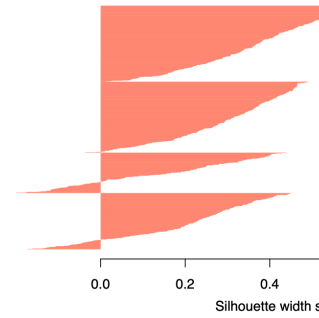
4 : 97 | 0.18

5 : 96 | 0.22

Average silhouette width : 0.29

Fuzzy K-Means Silhouette | k=4 | Mean/Mode Imputed

n = 792



4 clusters C_j

$j : n_j | \text{ave}_{i \in C_j} s_i$

1 : 247 | 0.36

2 : 231 | 0.30

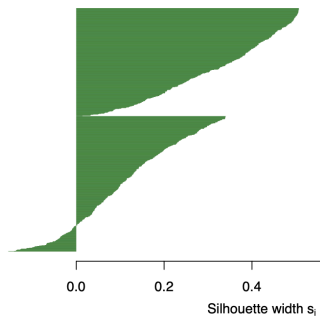
3 : 131 | 0.15

4 : 183 | 0.20

Average silhouette width : 0.27

PD Clustering Silhouette | k=2 | Mean/Mode Imputed

n = 792



2 clusters C_j

$j : n_j | \text{ave}_{i \in C_j} s_i$

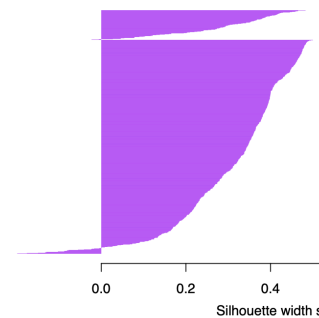
1 : 351 | 0.33

2 : 441 | 0.11

Average silhouette width : 0.21

Hierarchical Avg Silhouette | k=2 | Mean/Mode Imputed

n = 792



2 clusters C_j

$j : n_j | \text{ave}_{i \in C_j} s_i$

1 : 96 | 0.29

2 : 696 | 0.31

Average silhouette width : 0.31

EDA Tables:

Table 3: Descriptive Statistics — Continuous Variables

Variable	N	Mean	SD	Min	Q1	Median	Q3	Max	NA
amount	792	0.332	0.373	0.0	0.000	0.200	0.600	1	0
multiplier	792	0.990	0.585	0.0	0.480	1.005	1.502	2	0
political_views	519	64.027	29.307	0.0	45.500	71.000	88.000	100	273
price	791	3.135	9.318	0.5	0.664	0.990	2.062	100	1
religious_views	519	33.592	35.968	0.0	0.000	18.000	71.000	100	273
work_time_followup	521	272.131	567.510	15.0	42.000	64.000	128.000	3157	271
work_time_main	792	260.341	320.766	3.0	87.000	130.000	261.250	1718	0

Table 4: Gender Distribution

Category	N	%
Female	373	47.1
Male	411	51.9
Other / Prefer not to state	8	1.0

Table 5: Age Distribution

Category	N	%
18-29	264	33.4
30-39	301	38.1
40-49	111	14.0
50-59	68	8.6
60-69	41	5.2
70 or over	6	0.8

Table 6: Income Distribution

Category	N	%
Under \$20,000	101	12.9
\$20,000 - \$39,999	190	24.3
\$40,000 - \$59,999	177	22.7
\$60,000 - \$79,999	130	16.6
\$80,000 - \$99,999	81	10.4
Over \$100,000	102	13.1

Table 7: 2016 Presidential Vote (Follow-up Respondents)

Category	N	%
Didn't vote	93	17.9
Donald Trump	157	30.3
Hillary Clinton	225	43.4
Someone else	44	8.5

Table 8: Political Party Affiliation (Follow-up Respondents)

Category	N	%
Democrat	238	45.9
Independent	143	27.6
Republican	127	24.5
Something else	11	2.1

Table 9: Religious Service Attendance (Follow-up Respondents)

Category	N	%
Never	241	56.2
A few times a year	75	17.5
Once or twice a month	39	9.1
Once a week	61	14.2
More than once a week	13	3.0

Table 10: Dictator Game Giving Behavior Summary

Statistic	Value
N	792.000
Mean Amount Sent	0.332
SD	0.373
Sent \$0 (N)	333.000
Sent \$0 (%)	42.000
Sent \$1.00 (N)	130.000
Sent \$1.00 (%)	16.400
Mean Given Gave > 0	0.573

Table 11: Distribution of Multiplier () — Binned in 0.25 Intervals

Multiplier Range	N	%
[0,0.25)	107	13.5
[0.25,0.5)	99	12.5
[0.5,0.75)	98	12.4
[0.75,1)	91	11.5
[1,1.25)	102	12.9
[1.25,1.5)	94	11.9
[1.5,1.75)	94	11.9
[1.75,2]	107	13.5

Table 12: Continuous Data ARI Comparisons

	MM GHD	Mice 1 Skew-t	Mice 1 GHD	Mice 2 Skew-t	Mice 2 GHD	KNN-5 Skew-t	KNN-5 GHD	KNN-10 Skew-t	KNN-10 GHD
MM GHD	1.000	0.234	0.079	0.208	0.064	0.229	0.086	0.237	0.074
Mice 1 Skew-t	0.234	1.000	0.125	0.591	0.043	0.657	0.062	0.704	0.045
Mice 1 GHD	0.079	0.125	1.000	0.095	0.629	0.093	0.727	0.093	0.660
Mice 2 Skew-t	0.208	0.591	0.095	1.000	0.087	0.607	0.066	0.587	0.048
Mice 2 GHD	0.064	0.043	0.629	0.087	1.000	0.044	0.654	0.042	0.681
KNN-5 Skew-t	0.229	0.657	0.093	0.607	0.044	1.000	0.090	0.765	0.046
KNN-5 GHD	0.086	0.062	0.727	0.066	0.654	0.090	1.000	0.071	0.802
KNN-10 Skew-t	0.237	0.704	0.093	0.587	0.042	0.765	0.071	1.000	0.058
KNN-10 GHD	0.074	0.045	0.660	0.048	0.681	0.046	0.802	0.058	1.000

Table 13: Continuous & Ordinal Data ARI Comparisons

	MM GHD	Mice 1 Skew-t	Mice 1 GHD	Mice 2 Skew-t	Mice 2 GHD	KNN-5 Skew-t	KNN-5 GHD	KNN-10 Skew-t	KNN-10 GHD
MM GHD	1.000	0.142	0.165	0.105	0.145	0.110	0.202	0.152	0.132
Mice 1 Skew-t	0.142	1.000	0.084	0.110	0.033	0.142	0.062	0.122	0.042
Mice 1 GHD	0.165	0.084	1.000	0.142	0.649	0.141	0.341	0.215	0.253
Mice 2 Skew-t	0.105	0.110	0.142	1.000	0.170	0.618	0.076	0.335	0.158
Mice 2 GHD	0.145	0.033	0.649	0.170	1.000	0.097	0.342	0.167	0.236
KNN-5 Skew-t	0.110	0.142	0.141	0.618	0.097	1.000	0.098	0.375	0.199
KNN-5 GHD	0.202	0.062	0.341	0.076	0.342	0.098	1.000	0.148	0.259
KNN-10 Skew-t	0.152	0.122	0.215	0.335	0.167	0.375	0.148	1.000	0.197
KNN-10 GHD	0.132	0.042	0.253	0.158	0.236	0.199	0.259	0.197	1.000